

# Mini-Project Proposal: AI Tutor with RAG and Conversational Learning

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### 1. Goals and Outcomes

Our project aims to develop an intelligent AI Tutor that provides personalized educational assistance through document-based question answering. The system will: (1) Answer questions from uploaded PDF course materials using Retrieval-Augmented Generation (RAG), (2) Generate contextual quiz questions to test understanding, (3) Maintain conversation context for coherent multi-turn dialogues, and (4) Adapt responses based on user comprehension. The deliverable is a web application that transforms static documents into an interactive learning companion.

### 2. Motivation

Traditional study methods involve passive reading without interactive feedback. Students struggle with complex concepts and lack immediate clarification. Our motivation is creating an active learning companion that engages students in Socratic dialogue, similar to human tutoring through targeted questions and explanations. This addresses the need for personalized educational AI that transforms static course materials into dynamic, interactive learning experiences accessible to any student.

### 3. Relevant Course Modules

This project directly applies and extends several course concepts: **Natural Language Processing** for text preprocessing, document chunking, and semantic understanding; **Question Answering Systems** through RAG architecture combining retrieval with generation; **Dialogue Systems** via conversation memory and multi-turn interaction management; **Machine Learning** using vector embeddings and similarity search for document retrieval; and **AI Tools** through integration of LLM frameworks and vector databases.

### 4. Technical Approach

We will use LangChain as the core framework with: **Document Processing** for PDF ingestion, intelligent chunking, and embedding generation using sentence transformers; **Vector Database** using ChromaDB for semantic similarity-based retrieval; **Conversational Agent** with LLM-powered tutoring prompts for adaptive questioning and explanations; **Memory System** implementing conversation buffers for context tracking; and **Interface** via Streamlit web application for intuitive chat interaction with document upload capabilities.

### 5. Feasibility and Timeline

This project is feasible within 2 weeks by leveraging existing frameworks rather than building from scratch. **Week 1:** Set up document ingestion pipeline and vector database, implement basic RAG functionality for Q&A, create initial tutor prompts and conversation logic. **Week 2:** Add quiz generation and adaptive features, implement conversation memory and context tracking, build Streamlit interface, conduct testing and debugging, complete final report. The scope is manageable using established tools (LangChain, ChromaDB, Streamlit) while meaningfully extending course concepts by combining document retrieval with conversational AI for educational applications.